



Inflation Rate

Lecture 10: Measuring the cost Of living

*A feast is made for laughter, and wine maketh merry
but money answereth all things." Ecl 10 v 19*



Outline



1. Measuring the Cost of Living
2. Inflation rate
 - Definitions
 - Effects of Inflation
 - Causes of Inflation
 - Inflation and Unemployment
 - Cures of Inflation

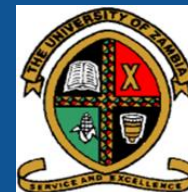


Measuring the cost of living



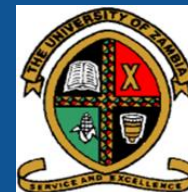
- People use their incomes to buy goods and services they need to live and enjoy life
- The price of goods and services determines how many commodities
- Overtime, price change and it becomes difficult to make comparisons about how wages and prices compare over time
- The cost of living- refer to how much money people need to maintain standards of living in terms of the goods and services they can afford
- Prices taken at a specific time give us the price level
- Price level – snapshot of the prices of goods and services in a particular economy at a particular period of time
- *The consumer price index (CPI) is a measure of the overall cost of the goods and services bought by a typical consumer*

Measuring the cost of living



- *Consumer price index (CPI)* – is a measure of the overall cost of the goods and services bought by a typical consumer
- Zamstats – collects the prices of selected goods and computes the CPI
- Steps to computing the Cpi
 - a) *Fix the Basket.* The first step in computing the consumer price index is to determine which prices are most important to the typical consumer
 - Assume typical HH buy hot dogs and than hamburgers
 - Also assume hh buys more hot dogs (4) than hamburgers (2) then the price of hot dogs is more important than the price of hamburgers and, therefore, should be given greater weight in measuring the cost of living
 - b) *Find the Prices.* The second step in computing the consumer price index is to find the prices of each of the goods and services in the basket for each point in time.

Measuring the cost of living



- c) **Compute the Basket's Cost.** The third step is to use the data on prices to calculate the cost of the basket of goods and services at different times.
 - We keep the basket of goods the same to isolate the effects of the price changes from the effect of any quantity changes
- d) *Choose a Base Year and Compute the Index.* The fourth step is to designate one year as the base year, which is the benchmark against which other years are compared
- e) Compute the Inflation Rate. The fifth and final step is to use the consumer price index to calculate the inflation rate

$$cpi = \frac{\text{price of the basket year}}{\text{Cost of the same basket in the base year}}$$

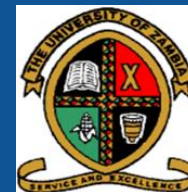
Compute the inflation rate

Problem in Measuring the cost of living



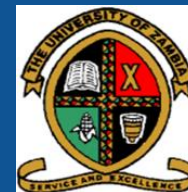
- The objectives of CPI is to measure changes in the cost of living
 - CPI tries to gauge how much incomes must rise in order to maintain a constant standard of living
 - The CPI index has three problems
1. Substitution bias – when prices change from one year to the next, they do not all change proportionately
- Some prices rise by more than others.
 - Consumers respond to these differing price changes by buying less of the goods whose prices have risen by large amounts and by buying more of the goods whose prices have risen less or perhaps even have fallen.
 - That is, consumers substitute toward goods that have become relatively less expensive
 - Yet the consumer price index is computed assuming a fixed basket of goods.
 - By not taking into account the possibility of consumer substitution, the index overstates the increase in the cost of living from one year to the next

Measuring the cost of living



2. *Introduction of new goods.* When a new good is introduced, consumers have more variety from which to choose
 - Greater variety, in turn, makes each kwacha more valuable, so consumers need fewer kwacha to maintain any given standard of living.
 - Yet because the consumer price index is based on a fixed basket of goods and services, it does not reflect this change in the purchasing power of the kwacha
3. *Unmeasured quality change – it does not take into account the changes in the quality. If the quality of a good deteriorates from year to the next, the effective value of the Kwacha falls, even if the price of the good stays the same*
 - Similarly, if the quality rises from one year to the next, the value of a kwacha rises & statisticians try to take the quality change in the price
 - By doing so they try to compute the price of a basket of goods of constant quality

Inflation



- Inflation is a phenomenon which affects everybody in one way or another
- Low levels or mild inflation in single digits (no more than 5% may be good for the economy
- However, inflation can have adverse effects on the economy
- It can lead or give rise to hyperinflation that is very rapidly accelerating inflation that leads to the breakdown of the monetary system. Example –Zimbabwe 2008

- What is inflation

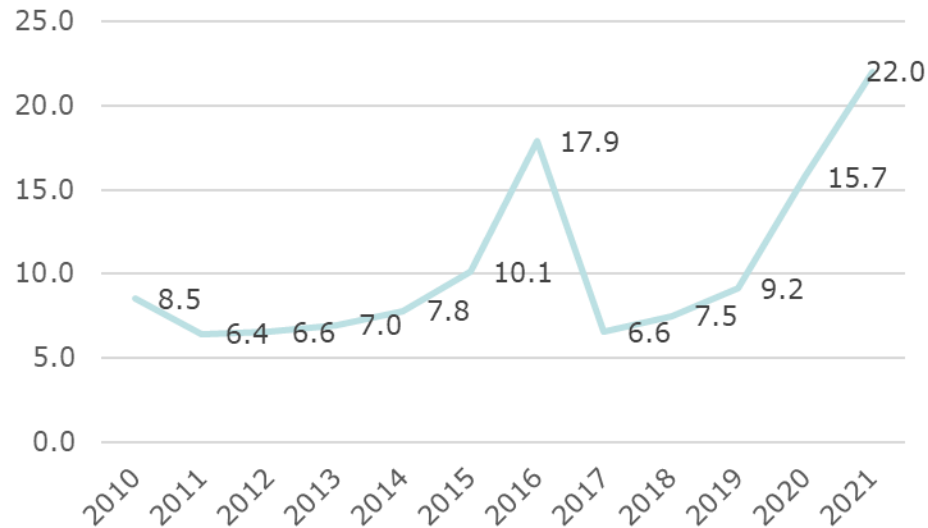


- Inflation is the persistent rise in the general price level. It only occurs when a large number of prices around the economy are rising together, creating an increase in the average price of the goods and services we buy
- inflation hurts the real buying of money

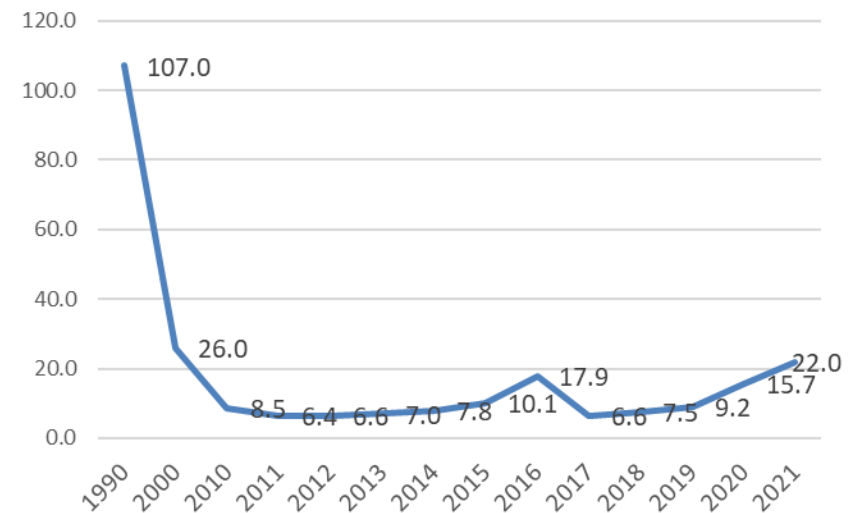
Inflation



Zambia's Inflation, consumer prices (annual %)



Zambia's Inflation, consumer prices (annual %)



Zambia experienced high and persistent rising prices all through the 1990s to mid-2000s

*Inflation dropped to single digit round 2007
However –rebounded in a fluctuating manner*

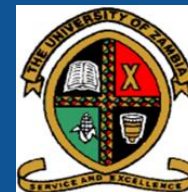
Inflation



Related concepts

- The persistence fall in the general or overall prices in the economy is called deflation
- As noted, measuring deflation and inflation is done by calculating the percentage change in the CPI ratio of the cost of the basket to the cost of that basket in the base year
- **Headline inflation** - -this the measure of all goods that the average consumer buys . It measures the changes in prices for the entire market basket of an average consumer
- **Core inflation** –is the measure of inflation that excludes goods with historical volatile price changes – energy and food

Costs/Effects of inflation



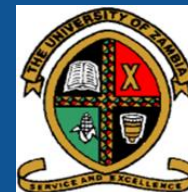
- Inflation can be anticipated or unanticipated
- If fully anticipated , then all groups of individuals in the economy expect it and are able to gain full compensation for it
- However, inflation may not be anticipated if:
 1. There is a general failure on part of the economy as whole to predict the inflation correctly so that the actual inflation exceeds the expected inflation
 2. Some groups or individuals in the economy fail to predict the inflation rate correctly so that they seek lower money wage increases than actually necessary to maintain the real wages
 3. Certain groups or individuals are unable to gain full compensation for it even though they anticipate it correctly

- **Costs of anticipated inflation**

1. *Shoeleather cost*

def: the costs and inconveniences of reducing money balances to avoid the inflation tax

Costs/Effects of inflation



1. *Shoe leather cost*

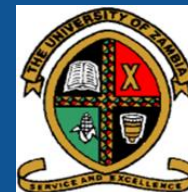
- Generally higher inflation rate leads to a higher nominal interest rate, which in turn leads to lower real money balances
- If people are to hold lower money balances on average, they must make more frequent trips to the bank to withdraw money
- The inconvenience of reducing money holding is metaphorically called the shoeleather cost of inflation, because walking to the bank more often causes one's shoes to wear out more quickly
 - **Recall even online transactions have costs of transactions –through charges**

2. *Menu costs*

- Second cost of inflation arises because high inflation induces firms to change
- their posted prices more often
- Changing prices is sometimes costly
 - **cost of printing new menus**
 - **cost of printing & mailing new catalogs**
 - **New Tv adverts etc**

The higher is inflation, the more frequently firms must change their prices and incur these menu costs

Costs/Effects of inflation



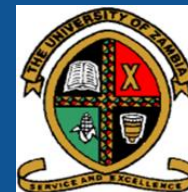
3. Relative price distortions

- A third cost of inflation arises because firms facing menu costs change prices infrequently;
 - therefore, the higher the rate of inflation, the greater the variability in relative prices
 - For example, suppose a firm issues a new catalog every January.
 - If there is no inflation, then the firm's prices relative to the overall price level are constant over the year
 - Yet if inflation is 1 percent per month, then from the beginning to the end of the year the firm's relative prices fall by 12 percent
 - Sales from this catalog will tend to be low early in the year (when its prices are relatively high) and high later in the year (when its prices are relatively low)
 - **causing microeconomic inefficiencies in the allocation of resources**

4. Unfair tax treatment

- Many provisions of the tax code do not take into account the effects of inflation.
- Inflation can alter individuals' tax liability, often in ways that lawmakers did not intend

Costs/Effects of inflation



4. *Unfair tax treatment*

- Inflation can alter individuals' tax liability, often in ways that lawmakers did not intend
- Ex. Suppose you buy some stock today and sell it a year from now at the same real price. It would seem reasonable for the government not to levy a tax, because you have earned no real income from this investment.
- Indeed, if there is no inflation, a zero tax liability would be the outcome
- But
 - suppose the rate of inflation is 12 percent and you initially paid K100 per share for the stock; for the real price to be the same a year later, you must sell the stock for **K112** per share.
 - In this case the tax code, which ignores the effects of inflation, says that you have earned K12 per share in income, and the government taxes you on this capital gain
 - The problem is that the tax code measures income as the nominal rather than the real capital gain

Costs/Effects of inflation

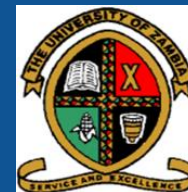


The Costs of Unexpected Inflation

- Unexpected inflation has an effect that is more pernicious than any of the costs of steady, anticipated inflation: it arbitrarily redistributes wealth among individuals
- Many long-term contracts not indexed but based on **inflation**. If **inflation** turns out different from expected, then some gain at others' expense.
- 1. **Unanticipated inflation hurts individuals on fixed- income earners and weakly unionized workers**
 - People with fixed income find out that the real value of their incomes become eroded by inflation
 - Weakly unionized workers who fail to bargain for full compensation for price rises will lose at the expense of strongly unionized workers
- 2. **From lenders to borrower** – lenders lose and borrower gain because when debt is repaid, their real value will be less than that prevailing when loans were made

Even when interest is paid, if it is less than inflation rate, there is a transfer of funds
- 4. **From Tax payers to government** - as income rises, earners with same real income move into higher tax bands and pay a bigger proportion of income in tax. This is called the fiscal drag and occurs in countries with a progressive tax system. Since government redistributes income to consumers, the final effect will be uncertain

Costs/Effects of inflation



5. From public sector to private sector – if government tries to control inflation using incomes and prices policy, it may set an example by resisting the wage claims by public sector workers.
 - If the private sector is more willing to concede wage increases, there will be redistribution from the public sector employees to the private sector employees. This depends on strength of unions and on ability of the private sector to provide wage increase
6. From profit earners to wage earners –if wage demands are met by squeezing profits margins, then the share of profits of the private sector can provide wage increases

Theories of the Causes of Inflation



- There are various schools of thought about the causes of inflation. The major ones are the classical school of thought and the Keynesian school of thought.

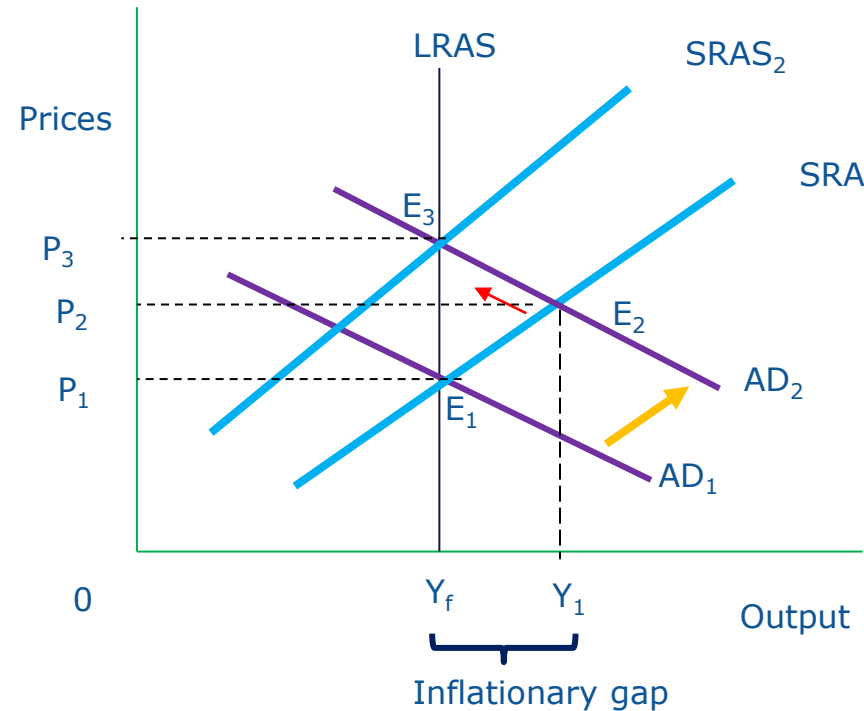
1) Demand Pull Inflation (Classical School of thought)

- This is based on the classical school of thought. They argue that *inflation is always and everywhere a monetary phenomenon*
- Demand-pull inflation is often summed up as “too many dollars chasing too few goods Or too much money chasing too few goods ”
- They argue that the problem is that we can’t produce any more goods because our economy is already operating at full capacity
- What happens next if demand keeps rising? What if people have money in their pockets and the desire to spend it?
- The aggregate demand will exceed aggregate output at (or close to) full employment
- Excess demand initiates inflationary pressure
- The excess demand can arise from the monetary (increase in money supply)or real sector (increase in G, I, C & NX)

Theories of the Causes of Inflation



- Assume economy is operating at full employment Y_f
- In short –run AD_1 and AS_1 are in equilibrium at E_1 which matches the long runs equilibrium
- Assume the central bank increased money supply
- The interest rate reduces, and investment rises which results in increased output
- AD increases due to increased money supply and shifts from AD_1 to AD_2
- A new At E_2 , the economy is not in equilibrium and the output is beyond the full employment level.
- Because demand exceeds supply, the price will begin to rise towards E_3 and firms will reduce production (why)
- AS shifts to AS_2

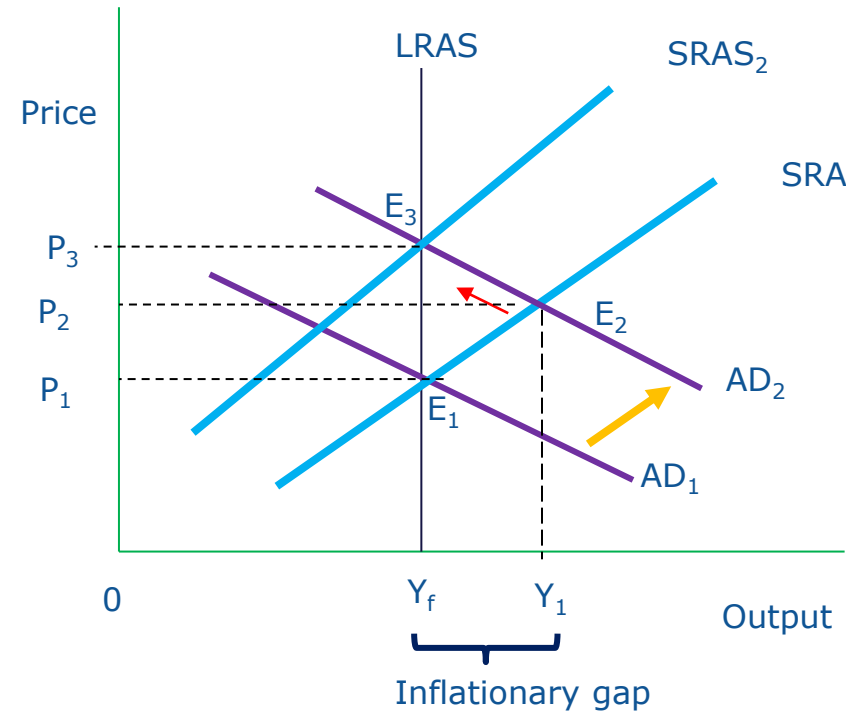


- Equilibrium is established at E_3 where the short-run AS and AD curves meet
- the result of an increase in money supply *is an increase in prices*

Theories of the Causes of Inflation



- This increase in money supply did not result in increased output
- Rather it resulted in the increase in nominal money supply and prices
- This is called the neutrality of money
- That is – the idea that in the long-run, changes in the money supply affect nominal variables such as prices and wages but do not affect the real outcomes in the economy

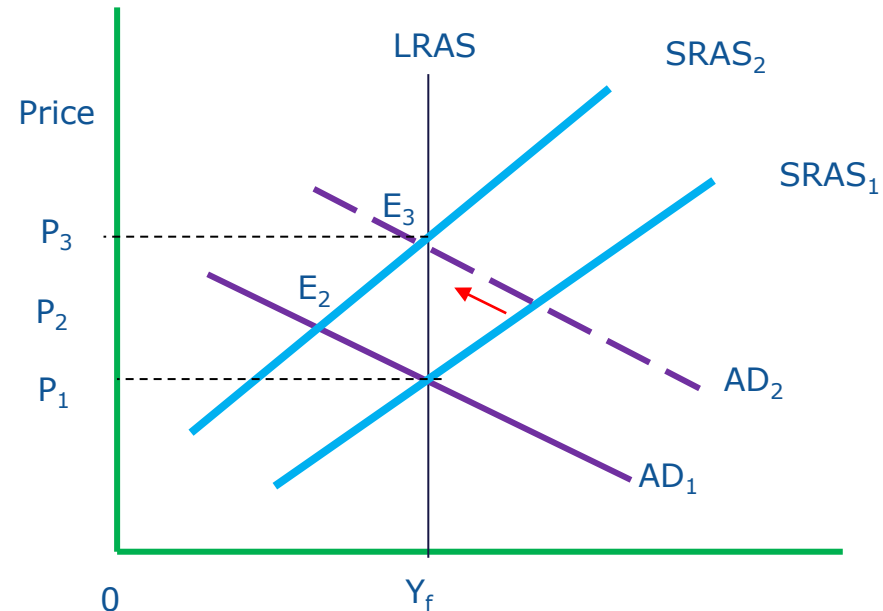


- The increase in money supply has resulted in increased prices.
- Hence- inflation is always and everywhere a monetary phenomenon



Cost Push Inflation

2. Cost push inflation arise due to rising money wages or other production costs are passed on to the consumer in form of higher prices
- ❑ As prices rise, real wages fall and workers demand higher wages and this increases the cost which is passed onto the prices
 - ❑ The continuous rise in the price and demand for wages results in the inflationary spiral
 - ❑ The increase in wages increases the cost of production and AS shifts to the left from $SRAS_1$ to $SRAS_2$ new equilibrium at E_2
 - ❑ Government may counter the decline in output by stimulating aggregate demand which shifts to the right
 - ❑ The economy finally settles at E_3 with same level of output but higher prices



Keynes argued that with unemployed resources, money wages would be more or less constant

At low levels of unemployment, money wages will begin to rise as the bottleneck occurred in the labour market

As full employment is reached, money wages rise faster.

Other causes of inflation



Imported inflation – this

occurs when price of imported final or intermediate goods rise

- The local price for the imported goods also go up to cover the costs. There are two reason imported prices rise:
- Rising world prices – if world prices are rising , all economies may not be spared by the inflationary pressure
- Currency depreciation – in a country with a flexible exchange rate like Zambia, the depreciation of the currency will through the pass-through effect raise domestic price of imported goods

Multi-causal

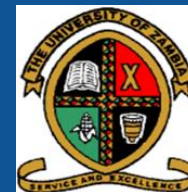
- *Excessive Government spending*
- *Excessive wage demands by unions*

Measures to address inflation

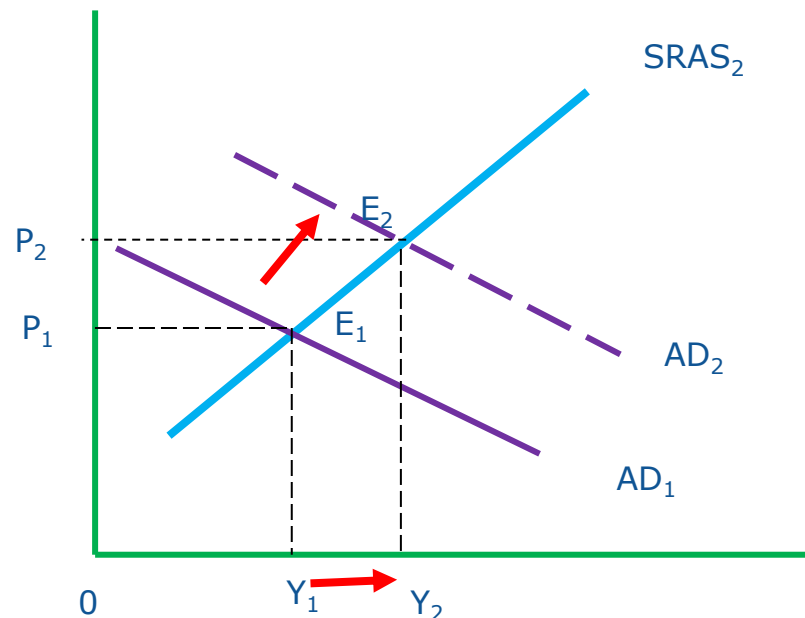


1. **Fiscal and Monetary policies** – commonly used to control inflation by affecting AD in the economy
 - Reduce inflation – cut G or I, increase taxes or some combination of these to reduce AD (deflationary fiscal policy)
 - A cut in Money supply or reduction in its rate of growth (deflationary monetary policy) will have the same effect
 - Keynesians – place emphasis on fiscal policy as a means of controlling AD while monetarists place a premium on monetary policy or money supply
2. **Prices and Incomes policy** – This is recommended by those who support the view that inflation is caused by unions demanding higher wages. It can take a number of approaches
 - *Government can encourage firms to avoid unjustified price rises and to unions to avoid unjustified wage claims*
 - *Bringing firms and unions together to obtain some voluntary agreement from both parties to keep prices and incomes down – this may require a norm for wage increases*
 - *Impose legislation regulate or even freeze wage and price freeze*
 - 1. *Indexation – is not a cure for inflation but rather a means of living with it. Adjust wages with increase in inflation*
 - *It can itself be inflationary*
 - *Does nothing to reduce other costs of inflation*

Inflation and Unemployment

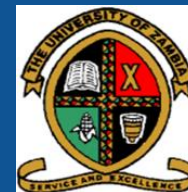


- In 1958, an economist named A.W. Phillips plotted the change in prices against unemployment over a 95-year period
- He observed the following:
 - As inflation ran high, unemployment was low
 - When inflation was low, unemployment was high
- The line showing this relationship is now called the Philips curve
- **Phillips Curve** – is a model that shows the connection between unemployment and inflation in the short-run
- *The relationship can be investigated using the aggregate demand and supply model*

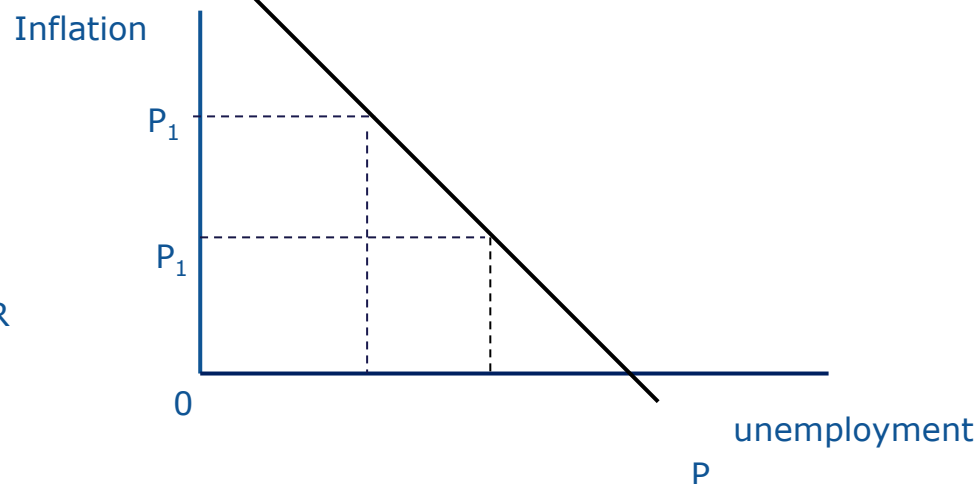
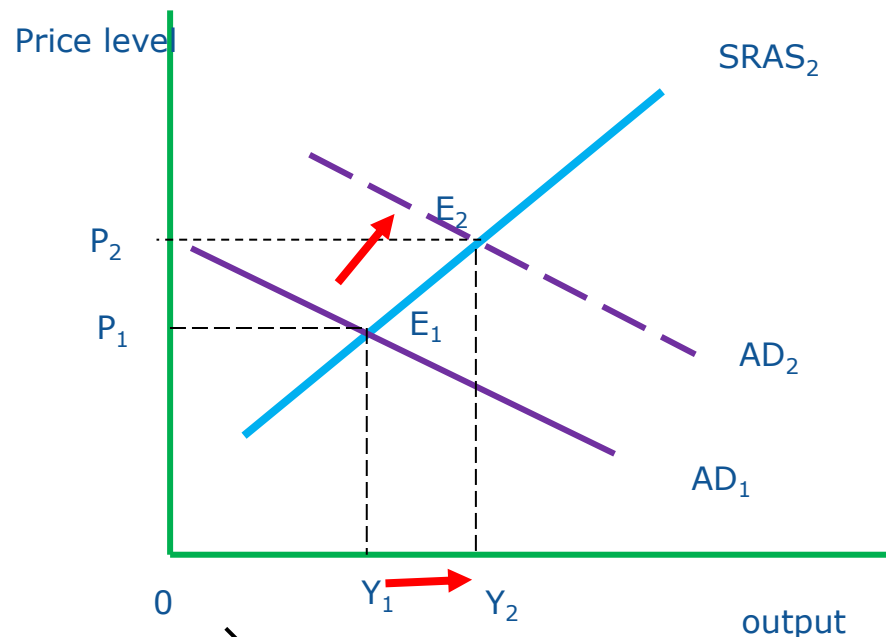


- *In the short-run, the economy is initially at E₁ with lower prices and out put than E₂*

Inflation and Unemployment



- With more economic activity, the aggregate demand curve moves to AD_2 and establishes a new equilibrium at E_2
- With higher output Y_2 , firms must employ more people
- At E_2 the prices are also higher.
- This relationship is plotted in the figure below
- An increase in AD above expectations will cause the economy to be at a point higher and to the left of the Philips curve



WE WILL DO THE DETAILS IN SECOND YEAR